



Pratham Chandratre

Most Asked Interview Questions for Data Scientist

Pratham Chandratre

Multi-Cloud Architect (7x Azure, 5x
Oracle, 2x GitHub, 2x Salesforce)





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1. Statistics and probability

- a. What is the difference between Type I and Type II errors?
- b. Explain p-value and its significance in hypothesis testing.
- c. What is the Central Limit Theorem, and why is it important?
- d. How do you handle an imbalanced dataset?
- e. What is multicollinearity, and how can you detect and fix it?





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2. Machine Learning

- a. What is the difference between supervised, unsupervised, and reinforcement learning?
- b. Explain bias-variance tradeoff with an example.
- c. What are precision, recall, and F1-score?
- d. When would you use them?
- e. What is overfitting? How can it be prevented?





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3. Deep Learning

- a. What is backpropagation, and how does it work?**
- b. How does batch normalization improve deep learning models?**
- c. What are the differences between LSTM and GRU?**
- d. Explain how attention mechanisms work in Transformers.**
- e. What is the difference between shallow and deep neural networks?**





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4. Feature Engineering

- a. What are some common data preprocessing techniques?
- b. How do you handle outliers in a dataset?
- c. Explain the difference between feature selection and feature extraction.
- d. What is the difference between PCA and t-SNE?
- e. How do you handle imbalanced datasets in classification problems?





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5. SQL & Databases

- a. Write an SQL query to find the second highest salary from an employee table.**
- b. What is the difference between UNION and UNION ALL?**
- c. Explain indexing in databases and its impact on query performance.**
- d. What is normalization in SQL? Explain different normal forms.**
- e. How would you find duplicate records in an SQL table?**





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6. Big Data & Cloud

- a. What is MapReduce, and how does it work in big data processing?**
- b. Explain the difference between Hadoop and Spark.**
- c. What are the advantages of using cloud storage in data science projects?**
- d. What are NoSQL databases, and when should you use them?**
- e. What is the main difference between structured and unstructured data?**





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